Practice Quiz: Planning in Robotic Systems

Part A: Multiple Choice

1. Planning in Active Inference is fundamentally about: A) Following a fixed pre-programmed schedule B) Evaluating the expected free energy of alternative action policies over future time steps and selecting the policy with the lowest expected free energy C) Acting randomly and hoping for the best D) Planning only for the next single time step

Answer: B -- Planning in Active Inference involves simulating multiple possible action sequences (policies) through the generative model, evaluating the expected free energy of each (combining pragmatic goal achievement with epistemic information gain), and selecting the policy that minimizes expected free energy over the planning horizon.

1. The expected free energy used for planning decomposes into: A) Hardware cost and software cost B) A pragmatic term (achieving preferred outcomes) and an epistemic term (reducing uncertainty about hidden states) C) Speed and accuracy only D) A single undivided quantity with no meaningful components

Answer: B -- This decomposition is fundamental to Active Inference planning. The pragmatic term drives the agent toward preferred states (reaching the goal). The epistemic term drives the agent toward informative states (reducing map uncertainty). Their balance determines whether the robot explores or exploits.

1. A mobile robot planning a path through an environment with known and unknown regions will: A) Always take the shortest path through known areas only B) Balance path efficiency (pragmatic value) with information gathering (epistemic value), potentially detouring through uncertain areas if the information gain is worth the extra travel C) Always explore unknown regions regardless of the mission D) Refuse to move until the entire map is known

Answer: B -- Expected free energy planning naturally trades off between reaching the goal efficiently and reducing map uncertainty. In high-uncertainty environments, the epistemic term can justify exploratory detours that improve the map, ultimately enabling more reliable future navigation.

1. Model Predictive Control (MPC) in robotics can be understood through Active Inference as: A) A completely unrelated approach B) An approximation to Active Inference planning where the generative model simulates dynamics over a finite horizon, evaluates action sequences against a cost function (analogous to expected free energy), and re-plans at each time step C) A method that only works for biological systems D) A planning method that does not use any model

Answer: B -- MPC and Active Inference planning share deep structural similarities: both use a forward model to simulate future states, both evaluate action sequences over a finite horizon, and both re-plan at each step as new observations arrive. The key difference is that Active Inference's expected free energy naturally includes both goal-seeking and uncertainty-reducing terms.

1. A manipulation robot planning a pick-and-place sequence for six objects must consider: A) Only the final arrangement, not the order B) The combinatorial space of possible orderings, physical constraints (which objects are accessible), and dependencies (which objects must be moved first) -- a task-level planning problem requiring a hierarchical generative model C) Each object independently with no consideration of other objects D) Only the lightest object first

Answer: B -- Task-level planning for multi-step manipulation requires reasoning about discrete state transitions (object locations, gripper state), geometric constraints (occlusion, reachability), and ordering dependencies. This demands a generative model that predicts task-state evolution under different action sequences.

1. The planning horizon -- how far into the future the agent simulates -- involves a trade-off between: A) Only computational cost, nothing else B) Planning quality (longer horizons yield better decisions) and computational cost (longer horizons require more simulation), with deeper horizons diminishing in value as prediction uncertainty compounds C) Color accuracy and processing speed D) Robot size and weight

Answer: B -- Prediction uncertainty grows with each simulated step, reducing the reliability of long-horizon predictions. The optimal planning horizon balances the value of longer foresight against both computational cost and increasing uncertainty. In Active Inference, this is naturally handled by precision decay over the planning horizon.

1. A robot using a behavior tree for task planning implements: A) A single reactive behavior with no planning B) A hierarchical, modular task plan where composite nodes (sequence, fallback, parallel) compose atomic action and condition nodes, enabling structured decision-making at the task level C) Only random action selection D) A flat list of actions with no structure

Answer: B -- Behavior trees provide a modular, hierarchical structure for task-level planning. From an Active Inference perspective, the behavior tree encodes the prior over policies -- the structured space of behavioral options that the agent evaluates when selecting actions.

1. Re-planning is triggered in Active Inference when: A) A fixed timer expires regardless of the situation B) Prediction errors exceed a threshold, indicating that the world has diverged from the planned trajectory and the current policy is no longer optimal C) The robot becomes bored D) Re-planning never occurs -- the initial plan is always sufficient

Answer: B -- When sensory observations significantly deviate from the predictions of the current plan, the plan's assumed future is no longer reliable. Re-planning generates a new policy from the updated belief state, ensuring the robot's actions remain appropriate for the actual world situation.

1. Hierarchical planning in robotics -- combining a task planner (which object to grasp) with a motion planner (how to reach it) -- corresponds to: A) Two unrelated computations B) Different levels of the generative model with different state spaces, timescales, and abstraction levels, where the task planner sets subgoals that constrain the motion planner C) Running the same algorithm twice D) Only using the motion planner and ignoring the task planner

Answer: B -- The task planner operates on a discrete state space (object locations, task states) at a slow timescale. The motion planner operates on a continuous configuration space (joint angles, end-effector poses) at a faster timescale. The task planner's output constrains the motion planner's goal, implementing hierarchical policy evaluation.

1. A drone planning a delivery route through an urban environment with no-fly zones and weather uncertainty must: A) Ignore all constraints and fly directly to the destination B) Evaluate policies that respect spatial constraints (no-fly zones), temporal constraints (delivery deadline), energy constraints (battery life), and uncertainty (weather forecasts with varying precision), selecting the policy with minimum expected free energy C) Wait indefinitely for perfect weather D) Only consider the shortest distance

Answer: B -- Real-world planning requires integrating multiple constraints into the generative model. Each constraint shapes the space of viable policies, and weather uncertainty introduces precision variations in the dynamics model. Active Inference planning naturally handles this by evaluating expected free energy under the full constrained, uncertain generative model.

Part B: Short Answer and Design Prompts

1. Design a planning system for a robot that must prepare a meal by coordinating multiple cooking steps with timing constraints (e.g., the pasta must finish at the same time as the sauce). How would the generative model represent temporal dependencies, and how would expected free energy evaluation select among alternative schedules?

2. A mobile robot must plan a path through a partially mapped building during an emergency evacuation. Some corridors may be blocked by debris. Describe how the Active Inference planning framework balances the pragmatic goal (reach the exit quickly) with the epistemic goal (determine which corridors are passable).

3. Compare RRT (Rapidly-exploring Random Tree) motion planning with Active Inference-based motion planning. What are the generative model equivalents of the RRT's sampling, nearest-neighbor, and steering operations?

4. A robot arm must plan a handover -- passing an object to a human at a mutually convenient location and timing. Design the planning component of the Active Inference model, including how the robot predicts the human's reach and adapts its plan to minimize mutual free energy.

5. How would you extend single-agent Active Inference planning to a multi-robot team that must collectively plan the exploration of an unknown building? Describe the shared generative model, the planning coordination mechanism, and how individual robot plans are reconciled.